

Chapter 3.1–3.3 - Fouriers Method I

3.2

Determine the coefficients in the following expansions

(a)

$$x = \sum_0^{\infty} \alpha_k \cos(k\pi x), \quad 0 < x < 1$$

(b)

$$x = \sum_1^{\infty} \alpha_k \sin(k\pi x), \quad 0 < x < 1$$

Solution:

(a)

$$\alpha_k = 2 \int_0^1 x \cos(k\pi x) \, dx = 2 \underbrace{\left[\frac{-x \sin(k\pi x)}{k\pi} \right]_0^1}_{=0} + 2 \int_0^1 \frac{\sin(k\pi x)}{k\pi} \, dx =$$

$$= \frac{2}{k\pi} \int_0^1 \sin(k\pi x) \, dx = \frac{2}{k\pi} \left[\frac{\cos(k\pi x)}{k\pi} \right]_0^1 = \frac{2 \cos(k\pi)}{k^2 \pi^2} - \frac{2}{k^2 \pi^2} =$$

$$= \frac{2}{k^2 \pi^2} (\cos(k\pi) - 1) = \frac{2}{k^2 \pi^2} ((-1)^k - 1)$$

$$\alpha_0 = \int_0^1 x \, dx = \frac{1}{2}$$

$$\alpha_k = \frac{2}{k^2 \pi^2} ((-1)^k - 1)$$

↓

$$x = \frac{1}{2} + \frac{2}{\pi^2} \sum_0^{\infty} \frac{(-1)^k - 1}{k^2} \cos(k\pi x)$$

(b)

$$\begin{aligned}
 \beta_k &= 2 \int_0^1 x \sin(k\pi x) \, dx = 2 \left[\frac{x \cos(k\pi x)}{k\pi} \right]_0^1 - \underbrace{2 \int_0^1 \frac{\cos(k\pi x)}{k\pi} \, dx}_{=0} = \\
 &= \frac{2 \cos(k\pi)}{k\pi} \\
 \beta_0 &= \int_0^1 x \sin(0\pi x) \, dx = 0 \\
 &\quad \downarrow \\
 x &= \frac{2}{\pi} \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} \sin(k\pi x)
 \end{aligned}$$

□

3A.5

Solve the heat problem

$$\begin{aligned}
 \frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} &= 0, & 0 < x < 1, t > 0 \\
 u(0, t) = u(1, t) &= 0, & t > 0 \\
 u(x, 0) &= \sin(\pi x) + 2 \sin(3\pi x), & 0 < x < 1
 \end{aligned}$$

Solution:

We have homogenous Dirichlet conditions, which should give our solution in the form of a sinus series. We can therefore express our solution in an infinite sum

$$u(x, t) = \sum_{k=1}^{\infty} u_k(t) \sin\left(\frac{k\pi x}{L}\right)$$

with $L = 1$. Using this expression we can differentiate termwise and put into the differential equation, resulting in

$$\begin{aligned}
& \frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} = 0 \\
& \downarrow \\
& \sum_{k=1}^{\infty} \underbrace{(u'_k(t) + k^2 \pi^2 u_{k(t)})}_{=0} \sin(k\pi x) = 0 \\
& \downarrow \\
& u'_k(t) + k^2 \pi^2 u_{k(t)} = 0 \\
& \downarrow \\
& u_k(t) = c_k e^{-k^2 \pi^2 t}
\end{aligned}$$

This gives our function $u(x, t)$ as

$$u(x, t) = \sum_{k=1}^{\infty} c_k e^{-k^2 \pi^2 t} \sin(k\pi x)$$

and using our initial conditions for $u(x, 0)$, we can determine the coefficients c_k :

$$\begin{aligned}
u(x, 0) &= \sum_{k=1}^{\infty} c_k \sin(k\pi x) = \sin(\pi x) + 2 \sin(3\pi x) \\
&\downarrow \\
c_k &= \begin{cases} 1, & k = 1 \\ 2, & k = 3 \\ 0, & \text{otherwise} \end{cases}
\end{aligned}$$

This gives the final solution

$$u(x, t) = e^{-\pi^2 t} \sin(k\pi x) + 2e^{-9\pi^2 t} \sin(3\pi x)$$

□

3A.7

A thin rod with closed ends has length $L = 1$. In the rod is air mixed with gas. Initially, the gas concentration is q in the left half of the rod, and 0 in the right half of the rod. Diffusion constant is set to $D = 1$. There is no source inside the rod. Write down a mathematical model and solve the problem.

Solution:

Using the diffusion equation

$$\begin{aligned}\frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} &= 0, & 0 < x < 1, t > 0 \\ \frac{\partial u(0, t)}{\partial x} &= \frac{\partial u(1, t)}{\partial x} = 0, & t > 0 \\ u(x, 0) &= \begin{cases} q, & 0 < x < \frac{1}{2} \\ 0, & \frac{1}{2} < x < 1 \end{cases}\end{aligned}$$

We have homogenous Neumann conditions, which will give the solution on the form of a cosine series expansion.

$$u(x, t) = \sum_{k=0}^{\infty} u_k(t) \cos(k\pi x)$$

Putting this into the differential equation and differentiating termwise gives

$$\begin{aligned}\frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} &= 0 \\ \downarrow \\ \sum_{k=0}^{\infty} (u'_k(t) + k^2 \pi^2 u_k(t)) \cos(k\pi x) &= 0 \\ \downarrow \\ u'_{k(t)} + k^2 \pi^2 u_k(t) &= 0 \\ \downarrow \\ u_k(t) &= c_k e^{-k^2 \pi^2 t}\end{aligned}$$

This gives the solution on the form

$$u(x, t) = \sum_{k=0}^{\infty} c_k e^{-k^2 \pi^2 t} \cos(k\pi x)$$

and we only need to determine the coefficients c_k , which is done by the initial value $u(x, 0)$.

We want to expand the function $u(x, 0)$ as a cosine series

$$\begin{aligned}u(x, 0) &= \sum_{k=0}^{\infty} c_k \cos(k\pi x) \\ \downarrow \\ c_k &= 2 \int_0^{\frac{1}{2}} q \cos(k\pi x) \, dx = 2 \left[\frac{q \sin(k\pi x)}{k\pi} \right]_0^{\frac{1}{2}} = \frac{2q \sin(\frac{k\pi}{2})}{k\pi} \\ c_0 &= \int_0^{\frac{1}{2}} q \, dx = \frac{q}{2}\end{aligned}$$

Final solution is then

$$u(x, t) = \frac{q}{2} + \frac{2q}{\pi} \sum_{k=1}^{\infty} \frac{\sin\left(\frac{k\pi}{2}\right)}{k} e^{-k^2 \pi^2 t} \cos(\pi k x)$$

□

3A.8

Solve the heat problem

$$\begin{aligned} \frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} &= 0, \quad 0 < x < 1, t > 0 \\ u(x, 0) &= x, \quad 0 < x < 1 \end{aligned}$$

with the boundary conditions

(a)

$$u(0, t) = u(1, t) = 0, \quad t > 0$$

(b)

$$\frac{\partial u(0, t)}{\partial x} = \frac{\partial u(1, t)}{\partial x} = 0, \quad t > 0$$

Solution:

(a) This problem has homogenous Dirichlet conditions, which will give the solution on sinus form.

$$u(x, t) = \sum_{k=1}^{\infty} u_k(t) \sin(k\pi x)$$

Putting this into the differential equation and differentiating termwise gives

$$\begin{aligned} \frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} &= 0 \\ \downarrow \\ \sum_{k=1}^{\infty} (u'_k(t) + k^2 \pi^2 u_k(t)) \sin(k\pi x) &= 0 \\ \downarrow \\ u'_k(t) + k^2 \pi^2 u_k(t) &= 0 \\ \downarrow \\ u_k(t) &= c_k e^{-k^2 \pi^2 t} \end{aligned}$$

This gives

$$u(x, t) = \sum_{k=1}^{\infty} c_k e^{-k^2 \pi^2 t} \sin(k\pi x)$$

and

$$u(x, 0) = x = \sum_{k=1}^{\infty} c_k \sin(k\pi x)$$

from which we can determine the coefficients c_k .

$$c_k = 2 \int_0^1 x \sin(k\pi x) dx = 2 \left[\frac{-x \cos(k\pi x)}{k\pi} \right]_0^1 + \underbrace{2 \int_0^1 \frac{\cos(k\pi x)}{k\pi} dx}_{=0}$$

$$c_k = \frac{-2 \cos(k\pi)}{k\pi} = \frac{2}{k\pi} (-1)^{k+1}$$

$$c_0 = \int_0^1 x \sin(k\pi 0) dx = 0$$

Final solution is

$$u(x, t) = \frac{2}{\pi} \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} e^{-k^2 \pi^2 t} \sin(k\pi x)$$

(b) This problem has homogenous Neumann conditions, which will give the solution on cosine form.

$$u(x, t) = \sum_{k=0}^{\infty} u_k(t) \cos(k\pi x)$$

Putting into differential equation

$$\sum_{k=0}^{\infty} (u'_k(t) + k^2 \pi^2 u_k(t)) \cos(k\pi x) = 0$$

↓

$$u'_k(t) + k^2 \pi^2 u_k(t) = 0$$

↓

$$u_k(t) = c_k e^{-k^2 \pi^2 t}$$

$$u(x, t) = \sum_{k=0}^{\infty} c_k e^{-k^2 \pi^2 t} \cos(k\pi x)$$

$$u(x, 0) = x = \sum_{k=0}^{\infty} c_k \cos(k\pi x)$$

From this we can determine the coefficients c_k

$$\begin{aligned}
 c_k &= 2 \int_0^1 x \cos(k\pi x) \, dx = -2 \int_0^1 \frac{\sin(k\pi x)}{k\pi} \, dx = \\
 &= 2 \left[\frac{\cos(k\pi x)}{k^2 \pi^2} \right]_0^1 = \frac{2 \cos(k\pi)}{k^2 \pi^2} - \frac{2}{k^2 \pi^2} = \frac{2}{k^2 \pi^2} ((-1)^k - 1) \\
 c_0 &= \int_0^1 x \cos(k\pi 0) \, dx = \frac{1}{2}
 \end{aligned}$$

This gives our final solution as

$$u(x, t) = \frac{1}{2} + \frac{2}{\pi^2} \sum_{k=1}^{\infty} \frac{(-1)^k - 1}{k^2} e^{-k^2 \pi^2 t} \cos(k\pi x)$$

□

3A.10

Solution:

□

3A.12

Solve the oscillation problem

$$\begin{aligned}
 \frac{\partial^2 u}{\partial t^2} + 2 \frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} &= 0, \quad 0 < x < 1, t > 0 \\
 u(0, t) = u(1, t) &= 0, \quad t > 0 \\
 u(x, 0) &= 1, \quad 0 < x < 1 \\
 \frac{\partial u(x, 0)}{\partial t} &= 0, \quad 0 < x < 1
 \end{aligned}$$

Solution:

We have homogenous Dirichlet conditions which will give the solution on sinus form

$$u(x, t) = \sum_{k=1}^{\infty} u_k(t) \sin(k\pi x)$$

Insert into differential equation and differentiate termwise

$$\sum_{k=1}^{\infty} (u_k''(t) + 2u_k'(t) + k^2\pi^2 u_k(t)) \sin(k\pi x) = 0$$

↓

$$u_k''(t) + 2u_k'(t) + k^2\pi^2 u_k(t)$$

Solving the characteristic equation with $u_k = re^t$

$$r^2 + 2r + k^2\pi^2 = 0$$

↓

$$r = -1 + \sqrt{1 - k^2\pi^2} = -1 \pm i\sqrt{k^2\pi^2 - 1} = -1 \pm i\omega_k$$

↓

$$u_k(t) = e^{-t}(a_k \cos(\omega_k t) + b_k \sin(\omega_k t))$$

Gives the solution as

$$u(x, t) = \sum_{k=1}^{\infty} e^{-t}(a_k \cos(\omega_k t) + b_k \sin(\omega_k t)) \sin(k\pi x)$$

$$u(x, 0) = 1 = \sum_{k=1}^{\infty} a_k \sin(k\pi x)$$

↓

$$a_k = 2 \int_0^1 \sin(k\pi x) dx = 2 \left[\frac{-\cos(k\pi x)}{k\pi} \right]_0^1 = \frac{-2 \cos(k\pi)}{k\pi} + \frac{2}{k\pi} \rightarrow a_k = \frac{2}{k\pi} (1 - (-1)^k)$$

$$\frac{\partial u(x, 0)}{\partial t} = 0 =$$

$$= \sum_{k=1}^{\infty} (-e^{-t}(a_k \cos(\omega_k t) + b_k \sin(\omega_k t)) + e^{-t}(-a_k \omega_k \sin(\omega_k t) + b_k \omega_k \cos(\omega_k t))) \sin(k\pi x) =$$

$$= \sum_{k=1}^{\infty} (-a_k + \omega_k b_k) \sin(k\pi x) = 0$$

↓

$$-a_k + \omega_k b_k = 0 \rightarrow b_k = \frac{a_k}{\omega_k} = \frac{2}{\omega_k k\pi} (1 - (-1)^k)$$

Full solution:

$$u(x, t) = \frac{2}{\pi} \sum_{k=1}^{\infty} e^{-t} \frac{1 - (-1)^k}{k} \left(\cos(\omega_k t) + \frac{\sin(\omega_k t)}{\omega_k} \right) \sin(k\pi x)$$

□

3A.13**Solution:**☐

3A.14**Solution:**☐

3A.15**Solution:**☐

3A.17**Solution:**☐

3A.19**Solution:**☐

3A.21**Solution:**☐

3A.22**Solution:**☐

3A.24a**Solution:**☐

3A.24bc**Solution:**☐

3A.25**Solution:**☐
